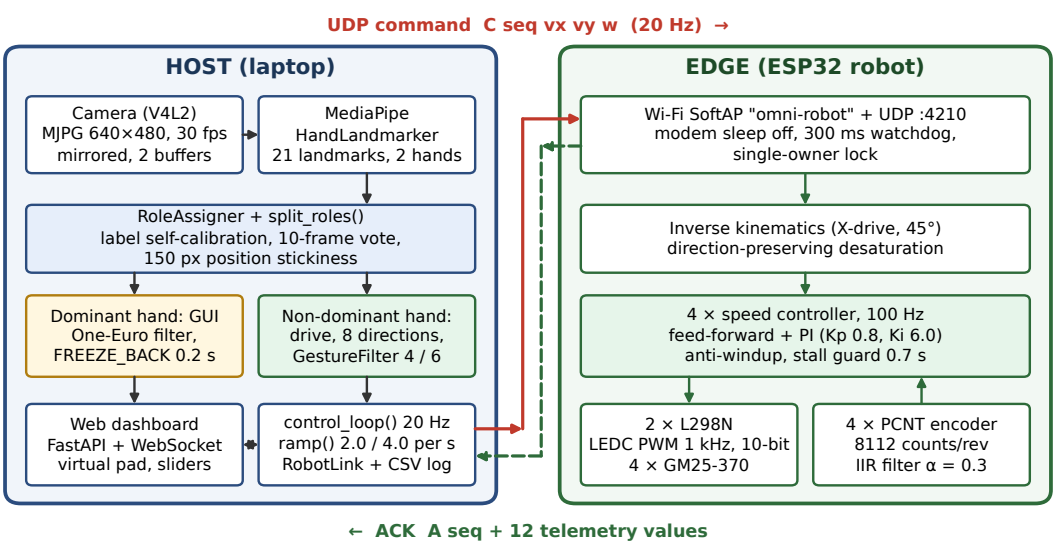
UDP command C seq vx vy w (20 Hz) →
HOST (laptop)
Camera (V4L2)
MJPG 640×480, 30 fps
mirrored, 2 buffers
MediaPipe
HandLandmarker
21 landmarks, 2 hands
RoleAssigner + split_roles()
label self-calibration, 10-frame vote,
150 px position stickiness
Dominant hand: GUI
One-Euro filter,
FREEZE_BACK 0.2 s
Non-dominant hand:
drive, 8 directions,
GestureFilter 4 / 6
Web dashboard
FastAPI + WebSocket
virtual pad, sliders
control_loop() 20 Hz
ramp() 2.0 / 4.0 per s
RobotLink + CSV log
EDGE (ESP32 robot)
Wi-Fi SoftAP "omni-robot" + UDP :4210
modem sleep off, 300 ms watchdog,
single-owner lock
Inverse kinematics (X-drive, 45°)
direction-preserving desaturation
4 × speed controller, 100 Hz
feed-forward + PI (Kp 0.8, Ki 6.0)
anti-windup, stall guard 0.7 s
2 × L298N
LEDC PWM 1 kHz, 10-bit
4 × GM25-370
4 × PCNT encoder
8112 counts/rev
IIR filter α = 0.3
← ACK A seq + 12 telemetry values